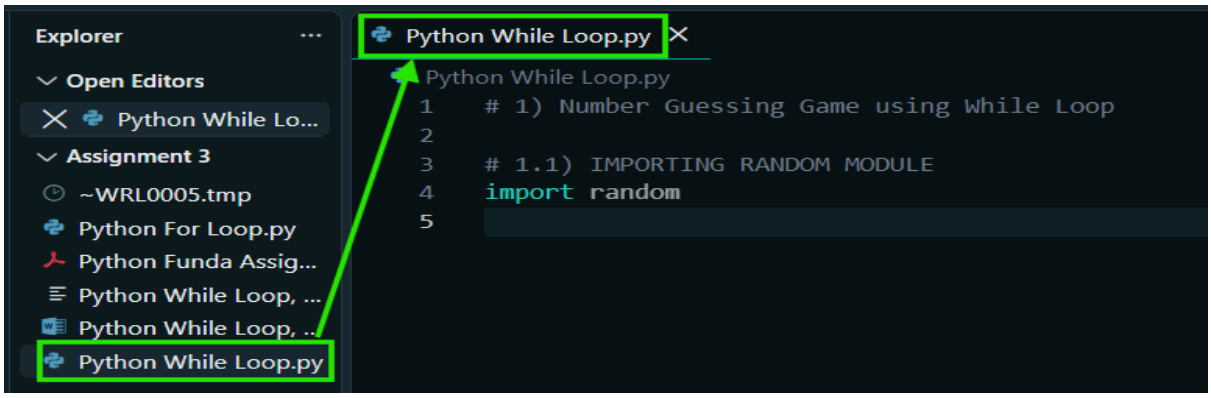
PYTHON ASSIGNMENT 3 – PYTHON

WHILE LOOP, FOR LOOP, FUNCTIONS

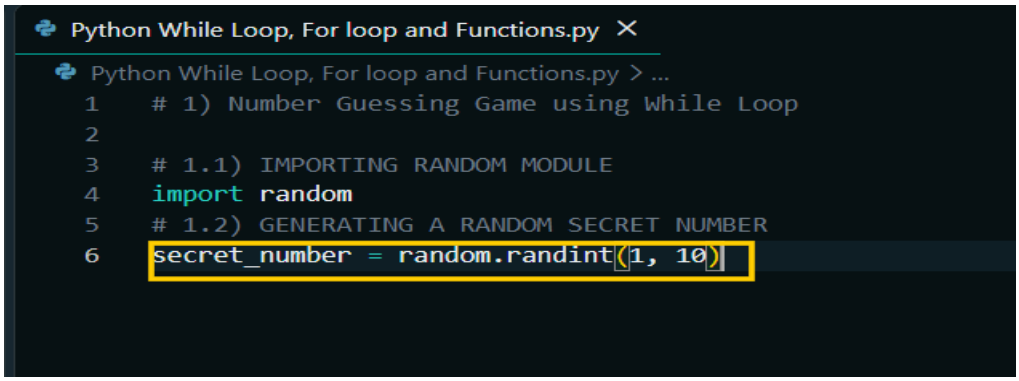
TOOLS: Python / VS Code
PROJECT TITLE: Python Loops and Functions – While Loop, For Loop, Functions
PROJECT OBJECTIVE: Working with Loops (while, for), Control Statements (break, continue, else) and Functions – understanding iteration, repetition, flow control and reusable code in Python programs
DATE OF SUBMISSION: 12/10/2026 – 1:30 PM

1) NUMBER GUESSING GAME USING WHILE LOOP:

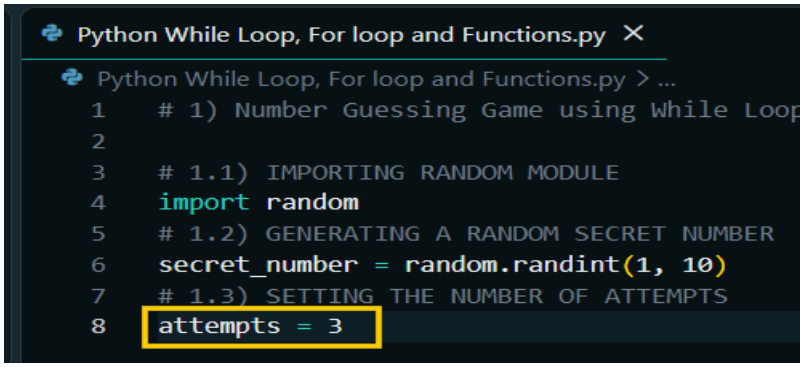
1.1) IMPORTING "RANDOM" MODULE IN THE NEW FILE "PYTHON WHILE LOOP.PY".



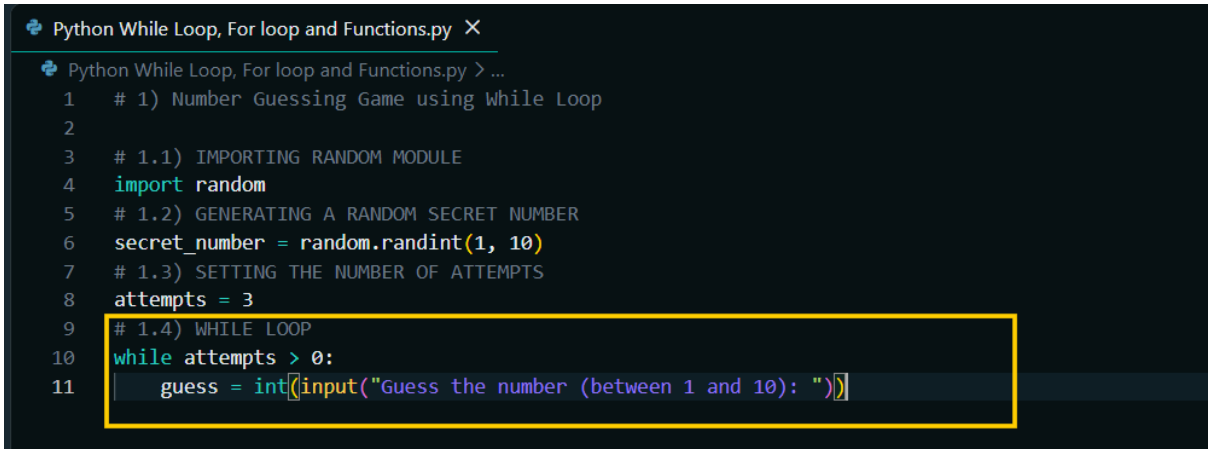
1.2) GENERATING A RANDOM SECRET NUMBER.



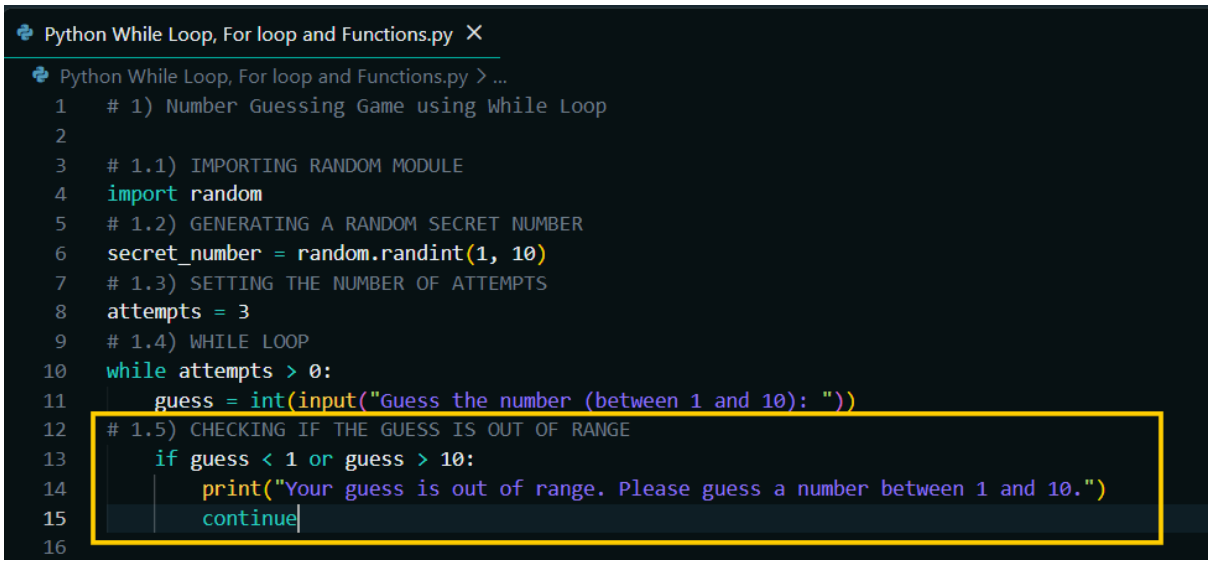
1.3) SETTING THE NUMBER OF ATTEMPTS.



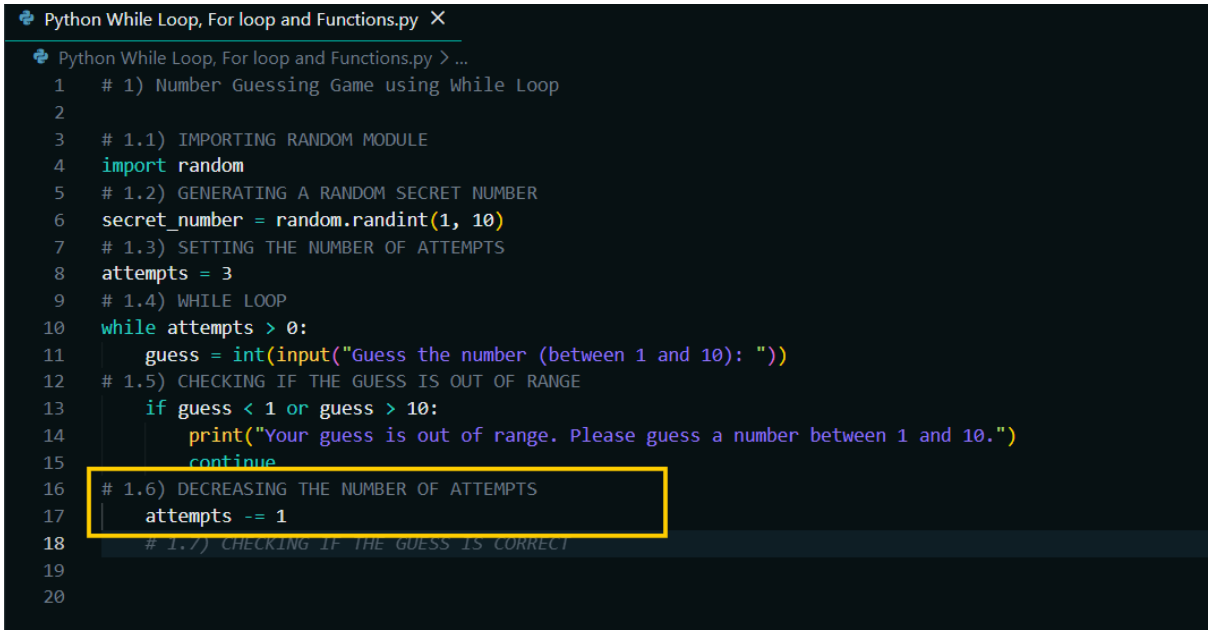
1.4) "WHILE" LOOP.



1.5) CHECKING IF THE GUESS IS OUT OF RANGE.



1.6) DECREASING THE NUMBER OF ATTEMPTS.

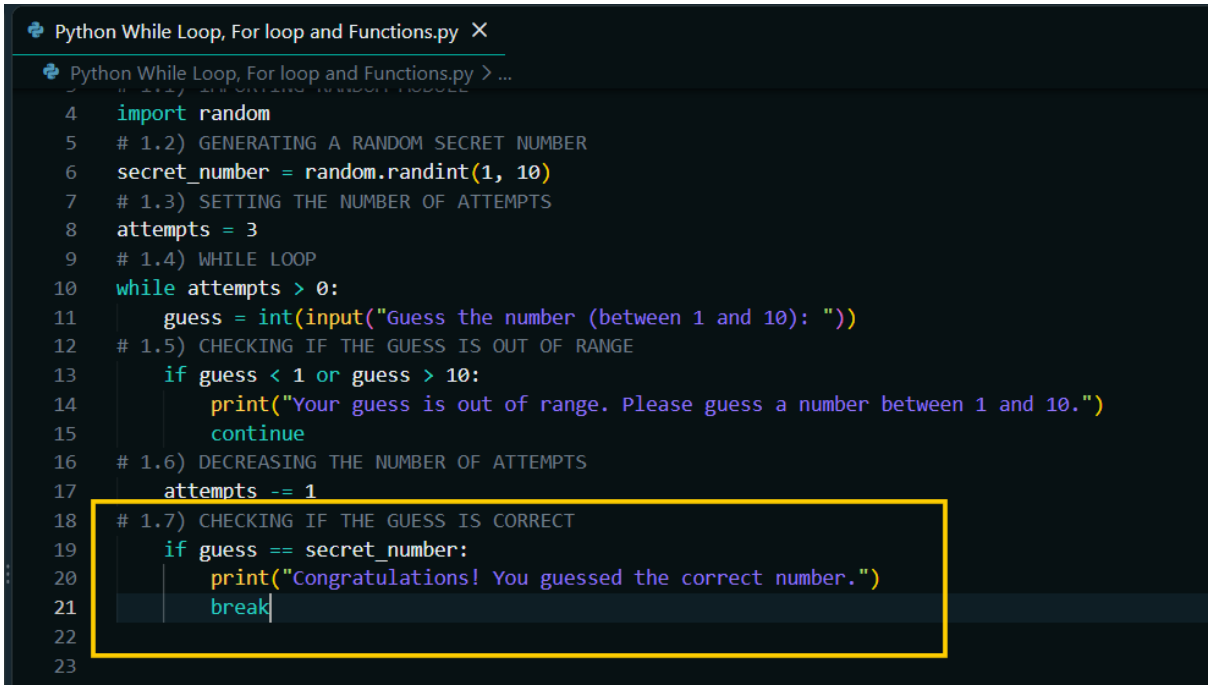


```python
# 1) Number Guessing Game using While Loop

# 1.1) IMPORTING RANDOM MODULE
import random
# 1.2) GENERATING A RANDOM SECRET NUMBER
secret_number = random.randint(1, 10)
# 1.3) SETTING THE NUMBER OF ATTEMPTS
attempts = 3
# 1.4) WHILE LOOP
while attempts > 0:
    guess = int(input("Guess the number (between 1 and 10): "))
# 1.5) CHECKING IF THE GUESS IS OUT OF RANGE
    if guess < 1 or guess > 10:
        print("Your guess is out of range. Please guess a number between 1 and 10.")
        continue
# 1.6) DECREASING THE NUMBER OF ATTEMPTS
    attempts -= 1
    # 1.7) CHECKING IF THE GUESS IS CORRECT


```

1.7) CHECKING IF THE GUESS IS CORRECT.



```python
import random
# 1.2) GENERATING A RANDOM SECRET NUMBER
secret_number = random.randint(1, 10)
# 1.3) SETTING THE NUMBER OF ATTEMPTS
attempts = 3
# 1.4) WHILE LOOP
while attempts > 0:
    guess = int(input("Guess the number (between 1 and 10): "))
# 1.5) CHECKING IF THE GUESS IS OUT OF RANGE
    if guess < 1 or guess > 10:
        print("Your guess is out of range. Please guess a number between 1 and 10.")
        continue
# 1.6) DECREASING THE NUMBER OF ATTEMPTS
    attempts -= 1
# 1.7) CHECKING IF THE GUESS IS CORRECT
    if guess == secret_number:
        print("Congratulations! You guessed the correct number.")
        break


```

1.8) CHECKING IF THE GUESS IS TOO LOW.



1.9) CHECKING IF THE GUESS IS TOO HIGH.

Python While Loop, For loop and Functions.py

```python
import random
# 1.2) GENERATING A RANDOM SECRET NUMBER
secret_number = random.randint(1, 10)
# 1.3) SETTING THE NUMBER OF ATTEMPTS
attempts = 3
# 1.4) WHILE LOOP
while attempts > 0:
    guess = int(input("Guess the number (between 1 and 10): "))
# 1.5) CHECKING IF THE GUESS IS OUT OF RANGE
    if guess < 1 or guess > 10:
        print("Your guess is out of range. Please guess a number between 1 and 10.")
        continue
# 1.6) DECREASING THE NUMBER OF ATTEMPTS
    attempts -= 1
# 1.7) CHECKING IF THE GUESS IS CORRECT
    if guess == secret_number:
        print("Congratulations! You guessed the correct number.")
        break
# 1.8) CHECKING IF THE GUESS IS TOO LOW
    elif guess < secret_number:
        print("Too Low. Try again.")

# 1.9) CHECKING IF THE GUESS IS TOO HIGH
    else:
        print("Too high. Try again.")

```

1.10) CHECKING IF THE USER HAS RUN OUT OF ATTEMPTS.

Python While Loop, For loop and Functions.py

```python
import random
# 1.2) GENERATING A RANDOM SECRET NUMBER
secret_number = random.randint(1, 10)
# 1.3) SETTING THE NUMBER OF ATTEMPTS
attempts = 3
# 1.4) WHILE LOOP
while attempts > 0:
    guess = int(input("Guess the number (between 1 and 10): "))
# 1.5) CHECKING IF THE GUESS IS OUT OF RANGE
    if guess < 1 or guess > 10:
        print("Your guess is out of range. Please guess a number between 1 and 10.")
        continue
# 1.6) DECREASING THE NUMBER OF ATTEMPTS
    attempts -= 1
# 1.7) CHECKING IF THE GUESS IS CORRECT
    if guess == secret_number:
        print("Congratulations! You guessed the correct number.")
        break
# 1.8) CHECKING IF THE GUESS IS TOO LOW
    elif guess < secret_number:
        print("Too Low. Try again.")

# 1.9) CHECKING IF THE GUESS IS TOO HIGH
    else:
        print("Too high. Try again.")
# 1.10) CHECKING IF THE USER HAS RUN OUT OF ATTEMPTS
else:
    print("Better luck next time!")
    print(f"The correct number was {secret_number}.")

```

COMPLETE CODE OF THE NUMBER GUESSING GAME USING "WHILE" LOOP.

```python
# 1) Number Guessing Game using While Loop

# 1.1) IMPORTING RANDOM MODULE
import random
# 1.2) GENERATING A RANDOM SECRET NUMBER
secret_number = random.randint(1, 10)
# 1.3) SETTING THE NUMBER OF ATTEMPTS
attempts = 3
# 1.4) WHILE LOOP
while attempts > 0:
    guess = int(input("Guess the number (between 1 and 10): "))
# 1.5) CHECKING IF THE GUESS IS OUT OF RANGE
    if guess < 1 or guess > 10:
        print("Your guess is out of range. Please guess a number between 1 and 10.")
        continue
# 1.6) DECREASING THE NUMBER OF ATTEMPTS
    attempts -= 1
# 1.7) CHECKING IF THE GUESS IS CORRECT
    if guess == secret_number:
        print("Congratulations! You guessed the correct number.")
        break
# 1.8) CHECKING IF THE GUESS IS TOO LOW
    elif guess < secret_number:
        print("Too Low. Try again.")

# 1.9) CHECKING IF THE GUESS IS TOO HIGH
    else:
        print("Too high. Try again.")
# 1.10) CHECKING IF THE USER HAS RUN OUT OF ATTEMPTS
else:
    print("Better luck next time!")
    print(f"The correct number was {secret_number}.")

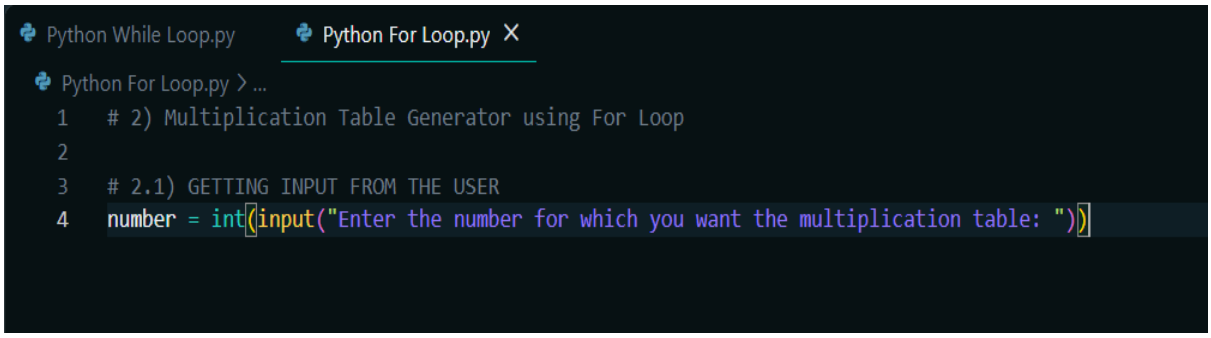
```

OUTPUT – NUMBER GUESSING GAME:

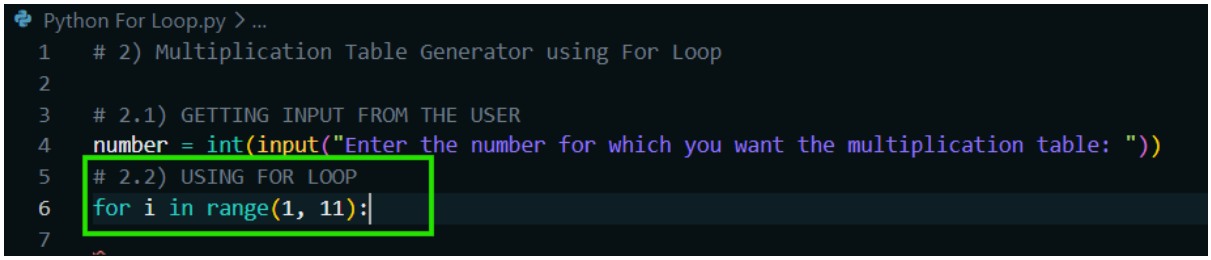
```
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 3> &
C:\Users\Venkadesh\AppData\Local\Programs\Python\Python314\python.exe
"d:/ENTRI ELEVATE DOCUMENTS/PYTHON PROGRAM/Assignment/Assignment 3/Python
While Loop.py"
Guess the number (between 1 and 10): 5
Too Low. Try again.
Guess the number (between 1 and 10): 7
Too Low. Try again.
Guess the number (between 1 and 10): 8
Congratulations! You guessed the correct number.
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 3>
```

2) MULTIPLICATION TABLE GENERATOR USING FOR LOOP:

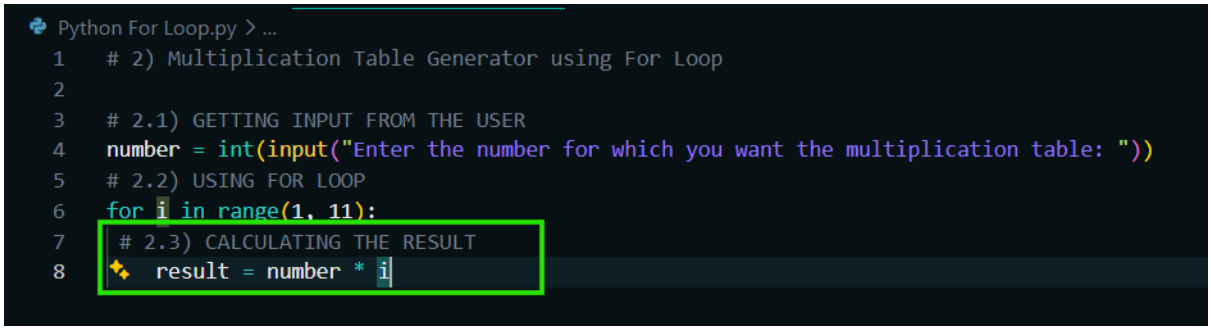
2.1) GETTING INPUT FROM THE USER IN THE NEW FILE "PYTHON FOR LOOP.PY".



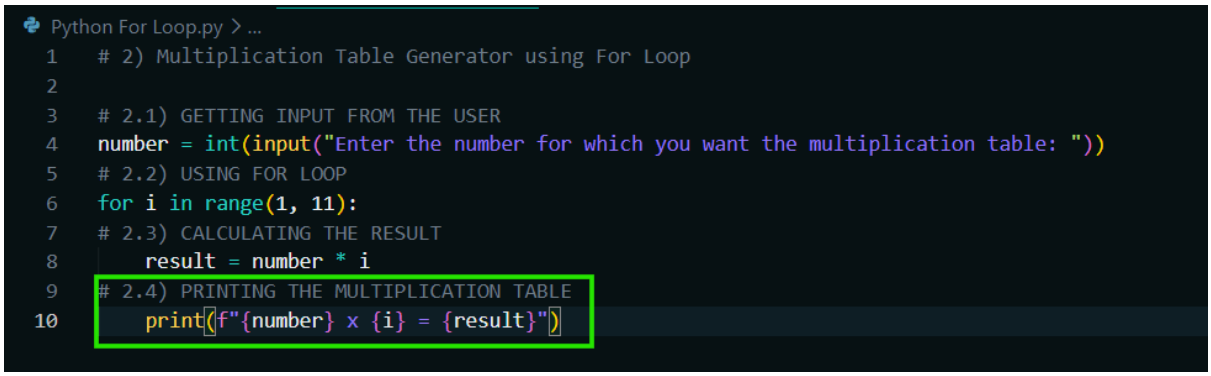
2.2) USING "FOR" LOOP.



2.3) CALCULATING THE RESULT.



2.4) PRINTING THE MULTIPLICATION TABLE.



COMPLETE CODE OF THE MULTIPLICATION TABLE GENERATOR USING "FOR" LOOP.

Python For Loop.py > ...

```python
# 2) Multiplication Table Generator using For Loop

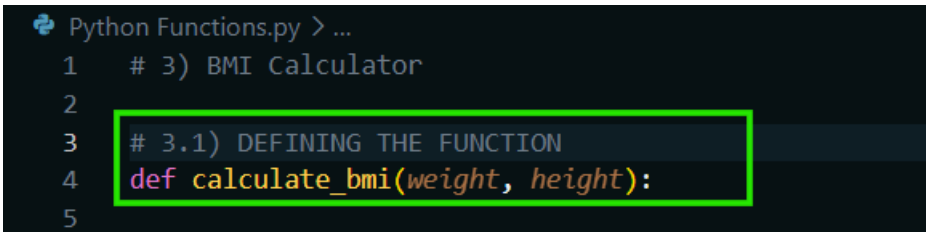
# 2.1) GETTING INPUT FROM THE USER
number = int(input("Enter the number for which you want the multiplication table: "))
# 2.2) USING FOR LOOP
for i in range(1, 11):
# 2.3) CALCULATING THE RESULT
    result = number * i
# 2.4) PRINTING THE MULTIPLICATION TABLE
    print(f"{number} x {i} = {result}")
```

OUTPUT – MULTIPLICATION TABLE:

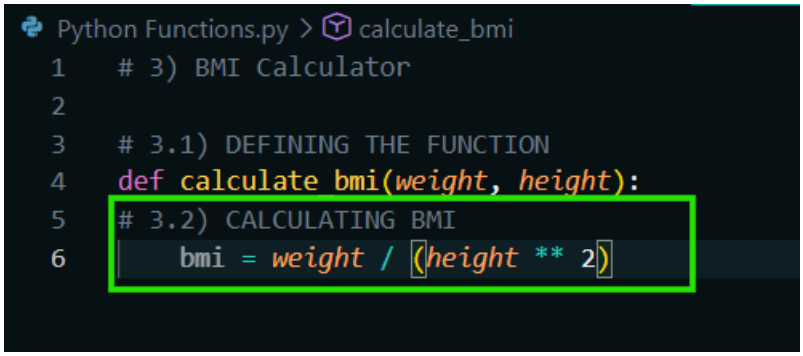
```
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 3> &
C:\Users\Venkadesh\AppData\Local\Programs\Python\Python314\python.exe
"d:/ENTRI ELEVATE DOCUMENTS/PYTHON PROGRAM/Assignment/Assignment 3/Python For
Loop.py"
Enter the number for which you want the multiplication table: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

3) BMI CALCULATOR USING FUNCTION:

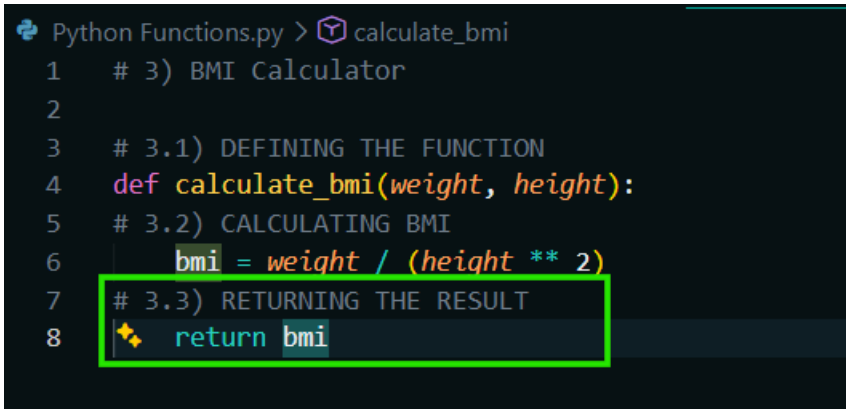
3.1) DEFINING THE FUNCTION IN THE NEW FILE "PYTHON FUNCTIONS.PY".



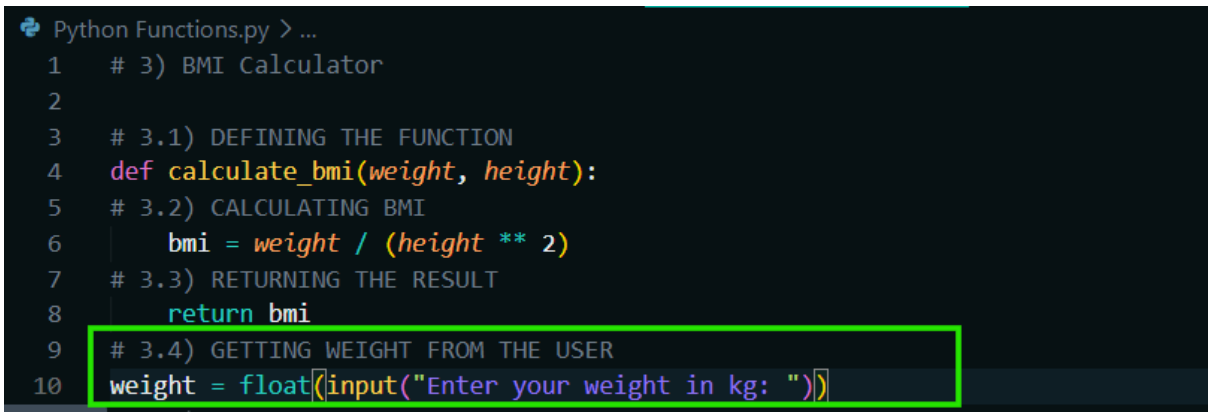
3.2) CALCULATING BMI.



3.3) RETURNING THE RESULT.



3.4) GETTING WEIGHT FROM THE USER.



3.5) GETTING HEIGHT FROM THE USER.

```python
# 3) BMI Calculator

# 3.1) DEFINING THE FUNCTION
def calculate_bmi(weight, height):
# 3.2) CALCULATING BMI
    bmi = weight / (height ** 2)
# 3.3) RETURNING THE RESULT
    return bmi
# 3.4) GETTING WEIGHT FROM THE USER
weight = float(input("Enter your weight in kg: "))
# 3.5) GETTING HEIGHT FROM THE USER
height = float(input("Enter your height in meters: "))
```

3.6) CALLING THE FUNCTION.

```python
# 3) BMI Calculator

# 3.1) DEFINING THE FUNCTION
def calculate_bmi(weight, height):
# 3.2) CALCULATING BMI
    bmi = weight / (height ** 2)
# 3.3) RETURNING THE RESULT
    return bmi
# 3.4) GETTING WEIGHT FROM THE USER
weight = float(input("Enter your weight in kg: "))
# 3.5) GETTING HEIGHT FROM THE USER
height = float(input("Enter your height in meters: "))
# 3.6) CALLING THE FUNCTION
bmi = calculate_bmi(weight, height)
```

3.7) DISPLAYING THE BMI.

```python
# 3) BMI Calculator

# 3.1) DEFINING THE FUNCTION
def calculate_bmi(weight, height):
# 3.2) CALCULATING BMI
    bmi = weight / (height ** 2)
# 3.3) RETURNING THE RESULT
    return bmi
# 3.4) GETTING WEIGHT FROM THE USER
weight = float(input("Enter your weight in kg: "))
# 3.5) GETTING HEIGHT FROM THE USER
height = float(input("Enter your height in meters: "))
# 3.6) CALLING THE FUNCTION
bmi = calculate_bmi(weight, height)
# 3.7) DISPLAYING THE BMI
print(f"Your BMI is: {bmi:.2f}")
```

SOURCE CODE – BMI CALCULATOR:

```python
# 3) BMI Calculator

# 3.1) DEFINING THE FUNCTION
def calculate_bmi(weight, height):
# 3.2) CALCULATING BMI
    bmi = weight / (height ** 2)
# 3.3) RETURNING THE RESULT
    return bmi
# 3.4) GETTING WEIGHT FROM THE USER
weight = float(input("Enter your weight in kg: "))
# 3.5) GETTING HEIGHT FROM THE USER
height = float(input("Enter your height in meters: "))
# 3.6) CALLING THE FUNCTION
bmi = calculate_bmi(weight, height)
# 3.7) DISPLAYING THE BMI
print(f"Your BMI is: {bmi:.2f}")
```

OUTPUT – BMI CALCULATOR:

```
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 3> &
C:\Users\Venkadesh\AppData\Local\Programs\Python\Python314\python.exe
"d:/ENTRI ELEVATE DOCUMENTS/PYTHON PROGRAM/Assignment/Assignment 3/Python
Functions.py"
Enter your weight in kg: 88
Enter your height in meters: 1.78
Your BMI is: 27.77
PS D:\ENTRI ELEVATE DOCUMENTS\PYTHON PROGRAM\Assignment\Assignment 3>
```